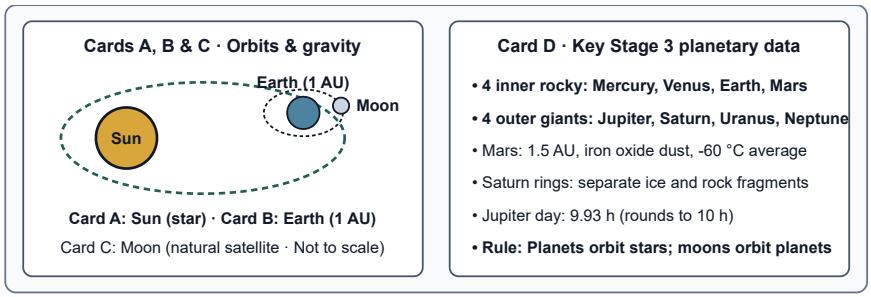
Reference sheets

Reference A · Class Solar System Source Cards (Cards A, B, C, and D)

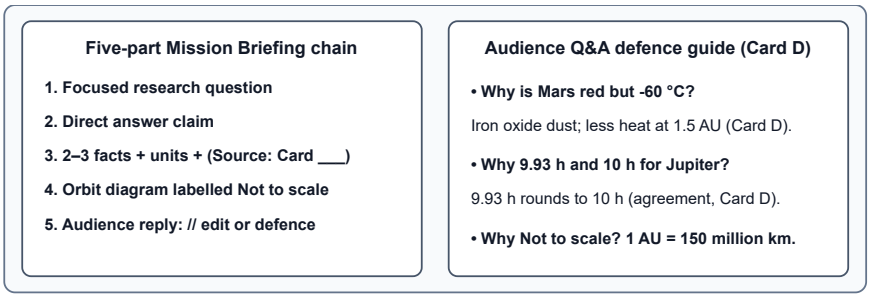


Class Source Cards A, B, C, and D showing the Sun's gravity, Earth's 1 AU orbit, the Moon's satellite orbit, and Key Stage 3 planetary data.

Reference only. Do not write on this sheet. Use these verified cards to build your Mission Briefing and answer audience questions.

Source card code	Body and classification	Verified science facts on this card	Key Stage 3 mechanism or measurement
Card A · Star card	The Sun (central star)	The Sun is a hot star at the centre of our Solar System that emits its own light	The Sun's non-contact gravitational pull holds planets in curved orbits around it
Card B · Planet card	Earth (inner rocky planet)	Earth is a nearly spherical rocky planet that orbits the Sun once a year	Earth orbits at an average distance of 1 astronomical unit (1 AU, about 150 million km)
Card C · Moon card	The Moon (natural satellite)	The Moon is a rocky natural satellite that orbits Earth in about a month and reflects sunlight	Earth's gravity holds the Moon in orbit around Earth rather than directly around the Sun
Card D · Solar System data	4 inner rocky planets & 4 outer giants	Inner rocky: Mercury, Venus, Earth, Mars. Outer giants: Jupiter, Saturn, Uranus, Neptune	Mars (1.5 AU, -60 °C, iron oxide dust); Saturn rings (ice and rock fragments); Jupiter day (9.93 h, rounds to 10 h)

Reference B · Five-part Mission Briefing and audience Q&A guide



Five-part Mission Briefing chain (Question, Claim, Facts + Card codes, Not to scale diagram, Audience reply) and Key Stage 3 defence guide.

Briefing part	What to include in your presentation	Example from Cards A, B, C, and D
1. Research question	One focused question about a Solar System body or pattern	'Why is Mars colder than Earth even though Mars looks red?'
2. Answer claim	One direct scientific sentence that answers your question	'Mars is colder because it orbits further from the Sun, and its red colour comes from rusty surface dust.'
3. Supporting facts & codes	Two or three verified facts with units and (Source: Card ___)	'Earth orbits at 1 AU (Source: Card B); Mars orbits at 1.5 AU, averages -60 °C, and has iron oxide dust (Source: Card D).'
4. Diagram & scale note	A clear diagram of the bodies/orbits plus the words 'Not to scale'	'Sun, Earth (1 AU), and Mars (1.5 AU) orbit diagram — Not to scale (1 AU = 150 million km).'
5. Audience Q&A response	Answer a listener's question using a card code, then mark // or defend	'9.93 h and 10 h for Jupiter's day agree because 9.93 h rounds to 10 h (Source: Card D).'

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. From the arrival picture cue, explain how Step 1, Step 2, and Step 3 work together to make a scientific presentation trustworthy.

Help: Link source citation, model limitations, and peer review across the three panels.

2. Which non-contact force holds Earth in its 1 AU orbit in Step 2, and which card backs that force?

Help: Recall the Key Stage 3 force on Card A and the 1 AU orbit on Card B.

3. If an audience member asks why Mars (-60 °C at 1.5 AU) is colder than Earth (1 AU), how do orbital distance and solar heating explain this?

Help: Link distance in astronomical units (AU) to the spread of solar energy.

4. When responding to an audience challenge in Step 3, when should you defend your claim as it stands and when should you add a // revision?

Help: Compare a complete, well-cited claim with one missing a source code or unit.

You do tasks

1. Present a Key Stage 3 Mission Briefing comparing the four inner rocky planets with the four outer gas and ice giants, using gravity, orbital distance in AU (1 AU vs 1.5 AU), surface temperature (-60 °C), and iron oxide composition from Cards A, B, and D.

Staff I ☐ S ☐ C ☐

Help: Synthesise evidence from Card A (gravity), Card B (1 AU), and Card D (inner rocky vs outer giants, Mars at 1.5 AU and -60 °C) with exact citations.

2. An audience member challenges your data table: 'Source Table 1 says Jupiter's day is 9.93 hours, while Source Table 2 says 10 hours, and your diagram of theMoon's orbit is way too big compared with the Sun!' Defend both points using Key Stage 3 science and Card C / Card D. **Staff I ☐ S ☐ C ☐**

Help: Explain rounding (9.93 h to 10 h) and why classroom diagrams state 'Not to scale' for the Moon (natural satellite) and Sun.

3. Audit and rewrite a flawed peer presentation that states: 'Mars is a hot red star at 1.5 AU, and the Moon is a small planet orbiting Earth.' Mark // to show every correction and cite Cards A, B, C, and D. **Staff I ☐ S ☐ C ☐**

Help: Correct the star vs rocky planet error, the red/hot error, and the planet vs natural satellite error.

Optional participation-loop prompt: Present one part of your briefing to an adult, hear one audience question or feedback step, and respond below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Summarise how your briefing uses gravity, orbital distance (1 AU vs 1.5 AU), and body classification to distinguish stars, inner rocky planets, outer giants, and natural satellites.

2. How would you answer an audience member who claims that 9.93 hours and 10 hours for Jupiter's rotation prove the source cards are unreliable?

3. Why does scientific peer review require presenters to keep their original draft visible and mark revisions with // when responding to audience questions?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt